DSC5001: Statistical Machine Learning I

Week 1: Statistical Learning and Probability Foundations

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Semester A, 2026/27

What can we learn from data?

Before using formulas, start from three familiar questions.

Apartment prices

Given size, location, and age, how much might a new apartment cost?

Medical diagnosis

Given a patient's measurements, is the patient likely to have a disease?

Photo organization

Can we automatically group similar photos in a phone album?

Different data questions require different kinds of answers.

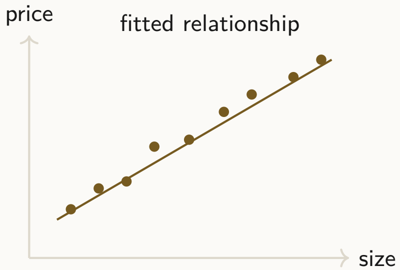
Regression: Predicting a Number

Question

Given observed features, predict a numerical value.

- **Input:** size, location, age, floor, distance to transit.
- **Output:** apartment price.
- **Basic idea:** learn how a numerical outcome changes with inputs.

Regression answers "how much?"



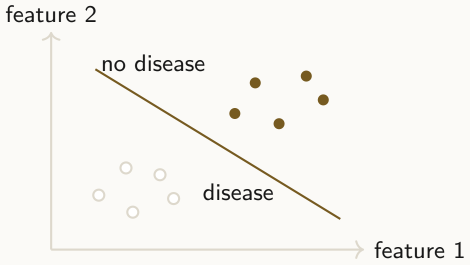
Classification: Choosing a Category

Question

Given observed features, predict a class or class probability.

- **Input:** lab measurements or image features.
- **Output:** disease / no disease.
- **Probability output:** "80% likely" is evidence, not a guarantee.

Classification answers "which category?"



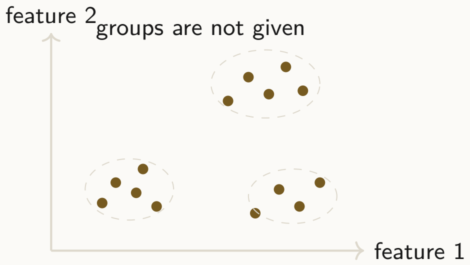
Clustering: Finding Structure without Labels

Question

Can we automatically group similar photos in a phone album?

- **Input:** photos represented by numerical features.
- **No labels:** nobody tells us which photos belong together.
- **Goal:** discover groups of similar photos automatically.

Clustering finds groups without predefined labels.



Three Learning Questions

Regression

Predict a numerical value. Example: apartment price.

Classification

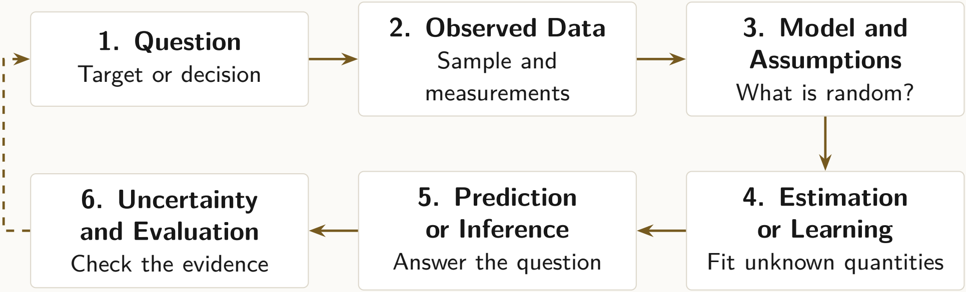
Predict a category or class probability. Example: disease status.

Clustering

Find groups without known labels. Example: photo groups.

The three problems ask for different kinds of answers.

A Statistical-Learning Workflow



Takeaway. Every method in this course occupies part of the same workflow.

Course Information

Item	Information
Instructor	Kangning Cui
Email	kangning.cui@cityu-dg.edu.cn
Office	AC4-422
Teaching Assistant	Yuanyuan Fang
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Lecture	Monday, 14:00–15:50
Tutorial	Monday, 16:00–16:50
Venue	AC5-515

Assessment

Assessment	Weight	Information
Assignments	10%	Two assignments covering theory, model analysis, and/or programming exercises.
Midterm Examination	20%	October 12 (Week 7).
Group Project	20%	Application of statistical machine learning methods to a real-world data problem.
Final Examination	50%	Final examination.

Detailed instructions will be posted on Canvas.

Roadmap I: Weeks 1–7

Tentative; pacing may be adjusted according to class progress

Week	Lecture Topic	Core Ideas	Tutorial / Activity
1	Introduction & Probability Review	Random variables; Bayes; moments; Gaussian	Probability practice
2	Statistical Inference I	Estimation; MoM; likelihood; MLE	MoM & MLE derivations
3	Statistical Inference II	Bootstrap; bias; variance; uncertainty	Bootstrap practice
4	Linear Regression	Least squares; multiple regression; assumptions	Regression fitting
5	Nonlinear Regression & Splines	Polynomials; basis functions; splines; bias–variance	Spline fitting
6	National Holiday / Revision Week	No class; midterm revision	Self-review
7	Midterm & Kernel Smoothing	Midterm; kernels; bandwidth; local regression	2-hour midterm + lecture

Roadmap II: Weeks 8–14

Tentative; pacing may be adjusted according to class progress

Week	Lecture Topic	Core Ideas	Tutorial / Activity
8	Gaussian Process Regression	GPs; covariance functions; prediction; uncertainty	GP prediction
9	Discriminant Analysis & Logistic Regression	LDA/QDA; logistic regression; odds	Classification practice
10	CART	Trees; splitting; impurity; pruning	Tree construction
11	Bagging, Random Forest & Boosting	Bagging; random forest; boosting	Ensemble comparison
12	Kernel Machines & SVM	Margin; regularization; kernels; SVM	SVM practice
13	PCA & Clustering	PCA; dimensionality reduction; K-means	PCA & clustering
14	Group Project Presentations	Presentations; demos; discussion	Project presentations

Learning objectives

1. Recognize regression, classification, and clustering questions.
2. Explain why probability is needed in statistical learning.
3. Use basic probability and Bayes' rule in a numerical example.
4. Distinguish population quantities from sample statistics.
5. Compute expectation and variance, and interpret covariance and Gaussian parameters.

Probability needed for learning

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Why probability is needed

- Two apartments with similar features can still have different prices.
- Similar medical measurements can still lead to uncertain diagnoses.
- The same set of photos may be grouped in more than one reasonable way.

Probability lets us describe uncertainty before we make a prediction, decision, or interpretation.

Why Statistics Still Matters

How this is used today

Modern systems are larger, but they still rely on probability models, estimation, optimization, validation, uncertainty, and careful interpretation.

The course develops reusable ideas through models whose assumptions and mathematics can be examined directly.

Probability and Distributions

Plain-language definition

Let Ω denote the set of all possible outcomes and A an event. Probability describes how likely an event is, while a probability distribution describes how uncertainty is spread over the possible values of a random quantity.

$$0 \leq \mathbb{P}(A) \leq 1, \qquad \mathbb{P}(\Omega) = 1$$

Main idea

Probability describes uncertainty before an outcome is known; a distribution organizes that uncertainty over all possible outcomes.

Random variables describe uncertain quantities

Plain-language definition

Suppose X is the number of customers who enter a shop in one hour. The count is not known in advance, so X is a random variable; its distribution describes which counts are common or rare.

$$X : \Omega \to \mathbb{R}, \qquad \omega \in \Omega, \qquad X(\omega) \in \{0, 1, 2, \ldots\}.$$

Main idea

Each $\omega \in \Omega$ is one possible outcome, and the function X assigns a numerical value to that outcome.

PMFs and PDFs

	Discrete variable	Continuous variable
What is assigned	probability to each value	density around each value
Notation	$p_X(x) = \mathbb{P}(X = x)$	$f_X(x)$
Total	$\sum_x p_X(x) = 1$	$\int f_X(x)\, dx = 1$
Example	number of successes	measured temperature

Common mistake

A probability density can exceed one; probability is obtained from area under the density curve.

Common Probability Models

Distribution	What it models	Example in this course
Bernoulli	one binary outcome	binary class label or coin flip
Binomial	number of successes	positives in repeated trials
Poisson	event count	arrivals during a fixed interval
Gaussian	continuous value near a center	measurement noise or class features

Conditional Probability

Plain-language definition

Conditional probability asks for the chance of one event after restricting attention to cases where another event occurred.

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0$$

Main idea

The denominator changes from the full population to the cases satisfying event B.

Probability connects the three learning questions

Learning question	Probability language	Meaning
Regression	$\mathbb{E}[Y \mid X = x]$ or $p(y \mid x)$	numerical outcome and uncertainty
Classification	$\mathbb{P}(Y = k \mid X = x)$	probability of each possible class
Clustering	structure in $p(x)$	patterns in observed data without known labels

These formulas now name the uncertainty behind the cases introduced at the start.

A Positive Test Changes Our Belief

Question

A disease is uncommon, but a patient's test is positive. How much should this result change our belief that the patient has the disease?

- We need to know how common the disease was before the test.
- We need to know how the test behaves with and without disease.
- We then update the probability after seeing the positive result.

Prior, Likelihood Terms, and Posterior

Prior

Before seeing the test result: $\mathbb{P}(D)$, the disease prevalence.

Likelihood terms

How probable the observed positive result is under each state: $\mathbb{P}(+ \mid D)$ and $\mathbb{P}(+ \mid D^c)$.

Posterior

After observing a positive test: $\mathbb{P}(D \mid +)$, the updated probability of disease.

Prior + observed evidence $\longrightarrow$ posterior belief.

Bayes' Rule for a Medical Test

$$\overbrace{\mathbb{P}(D \mid +)}^{\text{Posterior}} = \frac{\overbrace{\mathbb{P}(+ \mid D)}^{\text{Likelihood term}} \quad \overbrace{\mathbb{P}(D)}^{\text{Prior}}}{\mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c)}$$

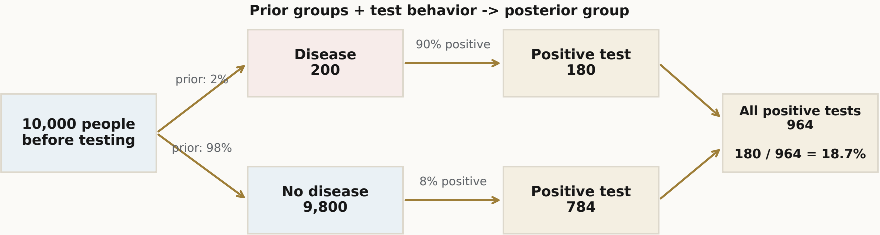
The denominator is the **evidence**:

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c).$$

$$\mathbb{P}(D \mid +) = \frac{0.90(0.02)}{0.90(0.02) + 0.08(0.98)} \approx 0.187.$$

A positive test updates the prior probability, while the denominator accounts for all ways a positive result can occur.

Medical-Test Counts



Before testing

200 have disease
9,800 do not

Positive tests

180 true positives
784 false positives

Posterior

$$\mathbb{P}(D \mid +) = \frac{180}{180 + 784} = 18.7\%$$

Takeaway. Among 964 positive tests, 180 come from people who actually have the disease.

Precision, Recall, and Specificity

Measure	**Meaning**	**Value**
Recall / sensitivity	Among actual positives, how many are detected? $\frac{TP}{TP+FN}$	$180/200 = 90\%$
Precision	Among predicted positives, how many are correct? $\frac{TP}{TP+FP}$	$180/964 = 18.7\%$
Specificity	Among actual negatives, how many are correctly rejected? $\frac{TN}{TN+FP}$	$9016/9800 = 92\%$

Recall/sensitivity finds positives; specificity rejects negatives; precision checks positive predictions.

Bayes' Rule for Parameter Inference

Data and parameter

Let $\mathcal{D}$ denote the observed data and θ an unknown model parameter. Here $p(\cdot)$ is used generically for a probability mass function or density, depending on the quantity.

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})} \propto p(\mathcal{D} \mid \theta)p(\theta).$$

Posterior = updated parameter distribution after observing the data.

Check conditional probabilities

Check your understanding. If sensitivity is 90%, does a positive test imply a 90% probability of disease? Explain which additional quantities are needed.

Write one sentence identifying the relevant definition or formula, then justify the answer.

Means, spread, and relationships

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A Fair Die

Bars

probability of each outcome

Equal heights

all outcomes are equally likely

Takeaway. A fair die assigns probability one sixth to every outcome.

Expected Value by Hand

Multiply each possible outcome by its probability, then add the six contributions.

$$\mathbb{E}[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \cdots + 6\left(\frac{1}{6}\right) = \frac{21}{6} = 3.5$$

Main idea

The expected value is the probability-weighted average across possible outcomes.

Common mistake

Do not divide by six again; the probabilities already supply the weights.

Can We Roll 3.5?

A single die roll can only be $1, 2, 3, 4, 5$, or 6.

No single roll equals 3.5. The value 3.5 describes the average over many rolls.

Expectation: A Long-Run Average

Plain-language definition

Expectation describes the long-run center of a random quantity under its probability distribution.

$$\mathbb{E}[X] = \sum_x x\, p_X(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx \quad \text{(continuous)}$$

Main idea

Use the probability mass function p_X for discrete values and the probability density f_X for continuous values.

Population Quantities and Sample Statistics

	Population or model	Observed sample
Mean	$\mu = \mathbb{E}[X]$	$\overline{x} = n^{-1} \sum_{i=1}^{n} x_i$
Variance	$\sigma^2 = \mathrm{Var}(X)$	$s^2 = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$
Role	fixed quantity, usually unknown	statistic that changes from sample to sample

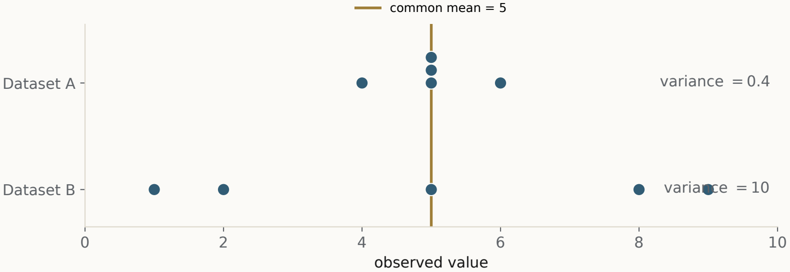
A sample statistic is evidence about a population quantity; the two are not interchangeable.

Two Datasets, Same Mean

$$
\begin{aligned}
A &= [4, 5, 5, 5, 6], & \qquad \overline{x}_A &= (4 + 5 + 5 + 5 + 6)/5 = 5, \\
B &= [1, 2, 5, 8, 9], & \qquad \overline{x}_B &= (1 + 2 + 5 + 8 + 9)/5 = 5.
\end{aligned}
$$

The mean alone cannot tell us how tightly the observations are grouped.

Which Dataset Is More Variable?



Dots
individual observations

Gold line
the common mean of five

Horizontal distance
deviation from the mean

Takeaway. Equal means can hide very different amounts of spread.

Variance by Hand

For these two listed datasets, compute empirical variance by averaging the five squared distances. This descriptive calculation is not the sample-variance estimator with denominator n-1.

$$v_{\text{emp}}(A) = \frac{1+0+0+0+1}{5} = 0.4, \qquad v_{\text{emp}}(B) = \frac{16+9+0+9+16}{5} = 10$$

Main idea

Dataset B has the larger variance because its squared distances from five are much larger.

Common mistake

Do not label these denominator-five summaries as s^2; s^2 uses denominator $n-1$ in this course.

Variance and Spread

Plain-language definition

Variance measures the average squared distance from the mean. Squaring prevents positive and negative deviations from cancelling.

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2], \qquad \mu = \mathbb{E}[X]$$

Main idea

Two datasets can have the same mean but very different variability because their values have different spread.

Variance Identity

Let $\mu = \mathbb{E}[X]$. Variance measures squared distance from the mean.

1. **Start from the definition.** $\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$
2. **Expand and use linearity of expectation.**

$$\mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2$$

3. **Use $\mathbb{E}[X] = \mu$.** $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Reading Covariance

Positive covariance

Study time and exam score often rise together.

Negative covariance

Temperature rises while heating demand falls.

Near-zero covariance

Two variables show no strong linear co-movement.

Covariance

Plain-language definition

Covariance describes how two variables vary together: it is positive when they tend to move in the same direction, negative when they tend to move in opposite directions, and near zero when there is little linear co-movement.

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

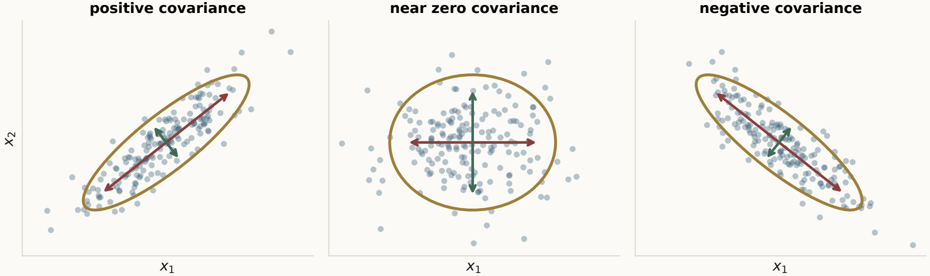
For two variables X and Y, their covariance matrix is

$$\mathbf{\Sigma} = \begin{pmatrix} \mathrm{Var}(X) & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \mathrm{Var}(Y) \end{pmatrix}.$$

Main idea

The diagonal entries measure the spread of each variable; the off-diagonal entries measure how the two variables move together.

Covariance Geometry



Dots

paired observations

Ellipse

main direction and spread

Slope

sign of covariance

Takeaway. Covariance sign follows whether the two variables move together or in opposite directions.

Gaussian models and preparation

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Gaussian Mean and Spread

Plain-language definition

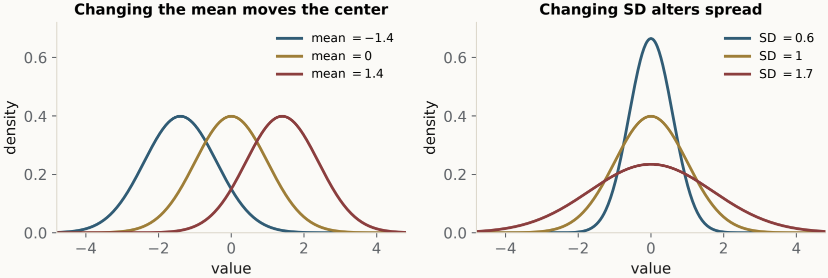
A Gaussian distribution is a smooth bell-shaped model. The mean locates its center and the variance controls its spread.

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

Main idea

μ and σ^2 are parameters: fixed but unknown quantities that describe the population model.

Gaussian Parameters



Peak location	Curve width	Area
the mean	the standard deviation	total probability equals one

Takeaway. The mean moves the Gaussian curve; the standard deviation changes its spread.

Multivariate Gaussian

Vector-valued data

When each observation contains several numerical features, we represent it by a random vector $\boldsymbol{X}$.

$$\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- $\boldsymbol{\mu}$: the mean vector, which determines the center.
- $\boldsymbol{\Sigma}$: the covariance matrix, which determines spread and orientation.

The one-dimensional ideas of mean and variance extend naturally to a mean vector and covariance matrix.

A Preview: Model Flexibility

Common mistake

Here, model variance refers to how a fitted model changes across different samples; it is different from the variance $\mathrm{Var}(X)$ of one random variable.

Bias–Variance View

More restrictive or smoother: A rigid probability model can miss real structure, producing systematic error.

More flexible or adaptive: An overly flexible model can follow random sample noise and change greatly across datasets.

The rest of the course develops methods for controlling this balance using model form, smoothing, averaging, and regularization.

Summary

- Regression predicts a number; classification predicts a category or class probability; clustering finds structure without labels.
- Probability describes uncertainty in data, predictions, and evidence.
- Bayes' rule combines prior information with observed evidence to update probability.
- Population quantities and sample statistics are related but not interchangeable.
- Expectation, variance, and covariance describe center, spread, and co-movement; Gaussian models use these quantities to describe distributions.

Week 1: connect the data question, mathematics, and evidence

Questions and discussion

Course Outcomes and Materials

Course outcomes

Explain statistical inference; build and evaluate regression and classification models; formulate regularized kernel methods; analyze bias and variance; and implement methods responsibly.

Materials

Slides, tutorials, code, and assignments will be posted on Canvas.

References

1. Casella and Berger, *Statistical Inference*.
2. Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning*.
3. Course lecture and tutorial notes.